

PROJECT

6.5

QUERY DATA WITH DYNAMODB

DATABASE SERIES

Prepared by:
Amrize JK
Certified Solutions Architect Associate

Objective

Learn to upload, query, and update data in DynamoDB using both the AWS Console and CloudShell. Practice real-time NoSQL operations in a serverless environment.

This project teaches how to handle multi-table updates with DynamoDB transactions just like in real-world cloud applications.

Services

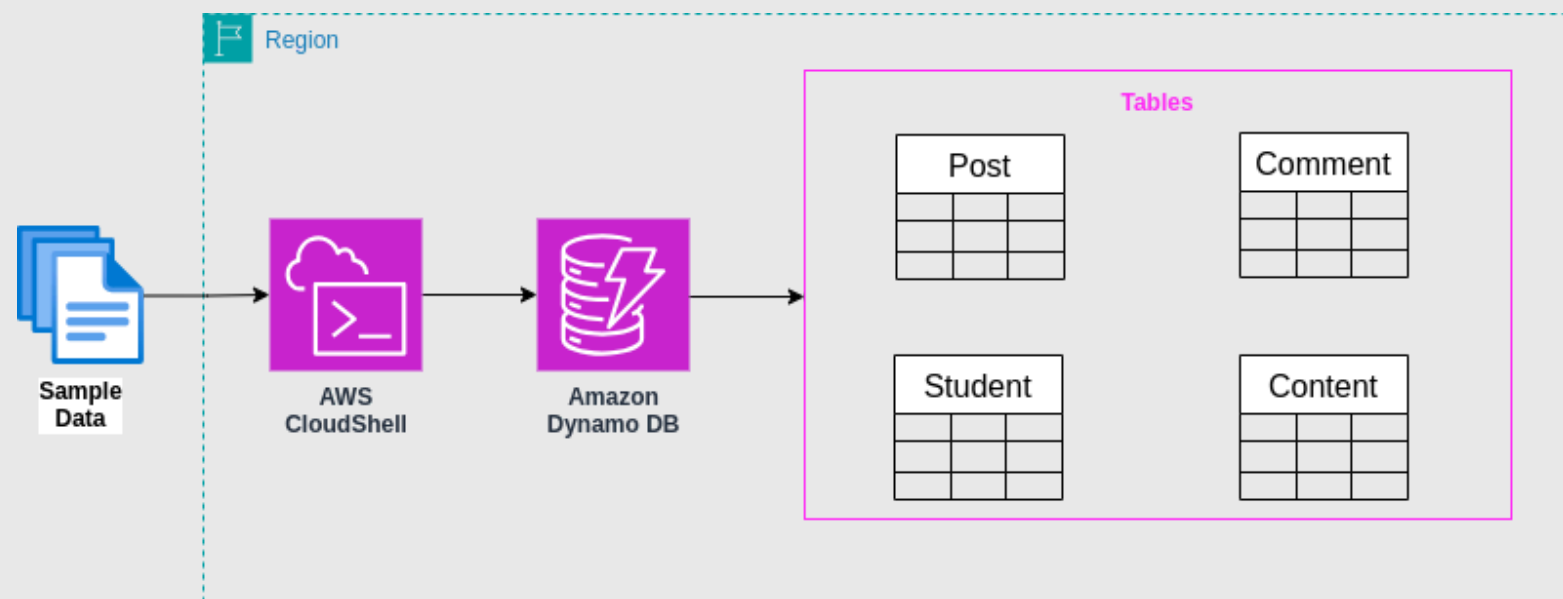
Amazon DynamoDB - A serverless NoSQL database to store and manage structured data.

AWS CloudShell - A browser-based CLI to interact with DynamoDB quickly.

AWS IAM - Manages access and permissions for DynamoDB operations.

AWS Management Console - A visual interface to create tables and run queries easily.

Architecture Diagram



This project runs entirely within AWS, where DynamoDB acts as your fast, serverless NoSQL database. You use CloudShell inside your VPC to create and query tables using the AWS CLI. IAM ensures secure access, while the console helps you interact visually. Transactions let you edit multiple tables in a single, atomic action all within a secure, managed AWS environment.

Steps to Implement 1

1. Login with IAM User:

Access the AWS Console using your IAM user credentials. This keeps your actions secure and gives the right level of permissions. Make sure you have access to DynamoDB and CloudShell.

2. Open AWS CloudShell:

Launch CloudShell from the AWS Console to begin working in a command-line environment. It's preconfigured with AWS CLI tools. No setup needed just open and you're ready.

3. Set Up DynamoDB Tables via CLI:

Use the AWS CLI in CloudShell to create DynamoDB tables. Define table names, primary keys, and any secondary indexes. This is the base structure for your data.

4. Verify Tables in Console:

Head over to the DynamoDB section in the AWS Console. Check that your tables were created successfully. It's a quick way to confirm before adding data.

Steps to Implement 2

5. Load Sample Data:

Insert data into your DynamoDB tables using AWS CLI put item or batch write item. Add multiple records to simulate real world data. This brings your tables to life!

6. Run Console Queries:

Use the AWS Console to run simple queries on your tables. Try searching for specific keys or attributes. This helps you understand how DynamoDB retrieves data.

7. Query Using CloudShell:

Go back to CloudShell and run the same queries using AWS CLI. Use get-item or query commands for practice. Now you've learned querying both ways!

8. Perform Transactions:

Use a DynamoDB transaction command to update two tables at once. Transactions ensure all actions succeed or none do. This step showcases real-world database consistency.

Story Time

DynamoDB (The Cloudy Treasure Vault):

DynamoDB is your magical vault where all your enchanted scrolls (data) are stored. It's super quick and doesn't need keys like traditional vaults it just needs the right secret code (partition key)! Ask it nicely, and it'll give you the treasure in a blink.

CloudShell (The Spell Room):

CloudShell is your cloud wizard's room always ready with scrolls and commands. Instead of wands, you type CLI spells like put-item and query to control your data vault, all from your browser, no setup needed!

AWS Console (The Oracle Mirror):

If you're not in the mood to type spells, use the Oracle Mirror also known as the AWS Console. Just click around to see your tables, add new items, or ask questions like, "What was the last potion brewed?"

IAM (The Access Wizard):

IAM is the wise access wizard that decides who gets to enter the spell room or open the treasure vault. It hands out permission scrolls to heroes like you, making sure no dark wizard gets in without approval!

Transactions (Twin Magic Scrolls):

Need to update two vaults at once? Use the powerful "transaction spell!" It's like sending two enchanted letters at the same time either both get delivered or none at all. That's true wizard grade reliability!

BY
AMRIZE J.K

THANK YOU!

